THERMAL WIND PREDICTION - DEFENSE NOTES & Q&A;

Key Message to Remember

Problem → Physical Hypothesis → Real Data → Validation → Machine Learning → Practical GO/NO GO Decision.

Question: Why is Accuracy lower than the Dummy Classifier?

The Dummy Classifier always predicts NO GO. Since roughly 69% of days do not have thermal wind, it achieves about 69% accuracy without learning anything. However, it detects zero good sailing days. My model sacrifices some accuracy in order to detect 61.6% of actual thermal wind events, which is the real objective.

Question: Why is Recall more important than Accuracy?

The dataset is imbalanced. Accuracy can be misleading. For a water sports user, missing a good thermal day is more costly than a false alarm. Therefore Recall, F1 Score and ROC-AUC are more informative metrics.

Question: Is the model good?

As a historical classifier, yes. It achieves ROC-AUC 0.714 and stable temporal validation. As an operational forecasting system, not yet. Future versions must incorporate forecast variables available before 07:00 and external weather models.

Question: What is the strongest result of the project?

Demonstrating that Sea Breeze occurrence is not random. The model learns physically meaningful mechanisms such as solar irradiance, thermal gradients and SST.

Final Reminder

If someone focuses only on Accuracy, explain the business objective. The goal is not to predict the majority class. The goal is to identify valuable GO days. The model catches more than 6 out of every 10 real thermal wind events.